GUNDA SAI VENKAT | ME17B013 P.R. No.: 22/ME/22/013

B.TECH.(HONS.) IN MECHANICAL ENGINEERING & M.TECH. IN PRODUCT DESIGN

EDUCATION AND SCHOLASTIC ACHIEVEMENTS

PROGRAM	INSTITUTION	%/CGPA	YEAR
Dual Degree in Mechanical Engineering	Indian Institute of Technology Madras	**9.22/10**	2022
XII (Telangana State Board)	FIITJEE Junior College, Hyderabad	**98.5%**	2017
X (Telangana State Board)	St. Paul's High School, Hyderabad	**10/10**	2015

- Selected for **RIYA, DAAD WISE and SFU MITACS** scholarships for summer of 2020 awarded to exceptional students for research internships.
- Recipient of the **Samsung IITM Pravartak Fellowship Award** offered to students working on exciting projects in **cyber-physical systems**.
- Pursuing **B.Tech Honors program**, awarded for excellent academic performance, involves taking **27 additional credits** of graduate courses.

PROFESSIONAL EXPERIENCE

RIYA Scholar, Ohio State University, USA
May 2020 - Aug' 2020

Dynamic Simulation of a Single Clearance Torsional System — ***Guide: Dr. Rajendra Singh***

- Performed **computational, damping and design studies** on a 2 DOF torsional system with a single clearance.
- Achieved **32 % reduction** in the **simulation time** by tuning and smoothening of simulation and solver parameters.
- Assessed role of **impact, hysteresis and viscous damping** and evaluated the respective damping coefficients.
- Identified **3 key design parameters** from a **set of 15** and optimized them to **reduce vibro-impacts by 28 %**.

PROJECTS

Dual Degree Project
May 2021 - Present

Acoustic Localization of Pipeline Inspection Robots — ***Guide: Dr. Prabhu Rajagopal***

- Developing a **unique acoustic system** for **accurate and real-time localization** of pipeline inspection robots.
- Identified **wave modes and frequencies** for fluid-pipe systems with the least attenuation, dispersion and noise.
- Conducted **FEA simulations** to evaluate the **range, signal quality and precision** of the guided acoustic waves.

Undergraduate Research Assistant
Sep' 2019 - Present

Cerebral Palsy Assist Wheelchair — ***Guide: Dr. Sujatha Srinivasan***

- Designed and fabricated **pediatric and adult electric wheelchairs** with **body movement-based control system**
- Wheelchairs useful for **cerebral palsy and quadriplegic patients** who are unable to control traditional joysticks.
- Incorporated **9 user-based customizations** to ensure postural support whilst enabling function and mobility.

Team Anveshak Centre for Innovation
Dec' 2017- May 2020

Semi-Autonomous All-Terrain Mars Rover — ***Guide: Dr. Asokan Thondiyath***

- Analyzed and developed **rocker-bogie suspension system** enabling rover to climb **45cm heights** and **30° slopes.**
- Guided the design of a **5 DOF serial manipulator** with a **compliant gripper** capable of lifting weights up to **5 kg**.
- Conceptualized a novel **automated science laboratory** to perform **in situ biological analysis** of soil and rocks.
- **Reduced 15% weight** through the use of **carbon fiber links** and **custom 3D printed TPU compliant wheels**.

Team Sahaay Centre for Innovation
Aug' 2017 – April 2019

Automated Waste Segregator

- Designed a **smart dustbin** that automatically segregates waste using **computer vision and machine learning**.
- Prototyped a **robust automated mechanism** to segregate and dispose waste accurately after identification.
- Achieved **90% accuracy** in waste detection and segregation in a **week-long user trial** conducted at IIT Madras.

Fall Protection Device

- Designed **wearable device** to detect person's fall and **inflate airbags** with **0.8 sec reaction time to prevent injury.**

Course Projects

- *Recommendation System for Bicycle Tour Prediction – CS5691*
- *Simulation of Battery Factory - ME3302*
- *Ultrasonic Detection Based Object Balancing Robots – ME2400*
- *Design of industrial gearbox - ME3301*

POSITIONS OF RESPONSIBILITY AND LEADERSHIP EXPERIENCE

Team Lead and Traversal Lead Team Anveshak
May 2019 – May 2020

- Spearheaded a **20-member multidisciplinary robotics team** across **3 modules** with a budget of **₹ 20 lakh.**
- Responsible for project division, planning, adherence to timeline, logistics, sponsorship and overall coordination.
- Collectively secured **sponsorship deals worth ₹ 8 lakhs** from IITM alumni, startups and tech companies.
- Initiated new strategies to **streamline the development process** for rapid prototyping and hassle-free integration.

Coordinator, CFI[1]
Aug' 2018 – April 2019

- Organized **workshops and summer school** for students to develop **frugal solutions** to **societal problems**.
- Coordinated **4 social innovation projects** with **15+ members** and oversaw their progress and expenditure

SKILLS

Software
- **Autodesk Fusion 360, Solidworks, Ansys, ABAQUS, Arduino, Aspen Plus, Plant Simulation, 3D Printing**

Programming
- **MATLAB & Simulink, Python (Numpy, Pandas, Scikit-learn), Mathematica, C/C++, LaTeX**

CO-CURRICULAR AND EXTRA CURRICULAR ACTIVITIES

International Rover Challenges
- Secured **8th** in **University Rover Challenge (URC) 2020**, **12th** in **URC 2019** out of **90+ international** teams.
- **Champions** of **Indian Rover Challenge 2019** beating the 2nd best team by a record margin of 55 points.

Inter IIT Tech Meet
- Placed **2nd** in **Campus Sustainability Challenge** in **Inter IIT Tech Meet 2018** conducted at IIT Bombay.

Other Competitions
- **1st place:** Shaastra Research Conference, 2018
- **1st place:** LitSoc[2] Cluedo 2019 and 2020.
- **2nd place:** Udhyam Innovation Challenge
- **Round 3:** Flipkart Grid 2.0 Robotics Challenge

Teaching

Volunteer | Speak Out | Bhumi NGO — ***Sep'2019 - Present***

- Teaching **English, communication and soft skills** every week for children from underprivileged communities.
- Awarded **Volunteer of the Month** for July 2020 for regular and enthusiastic contributions during lockdown.

Coding Mentor | CampK12 — ***Aug'2020 – Dec'2020***

- Trained children in app development, coding and logic skills with excellent student and parent feedback.

Camp Trainer | Young Leader's Initiative | KAARMIC Education — ***May 2018 - June 2018***

- Organized a **15-day camp** for English, extracurricular and soft skills in a **rural college for backward classes**.
- Trained **300+ students** in **basic computer skills** and assisted them in creating their own websites and blogs.

[1] Team Sahaay, Centre for Innovation

[2] LitSoc - Inter-Hostel Literary and Cultural Competition